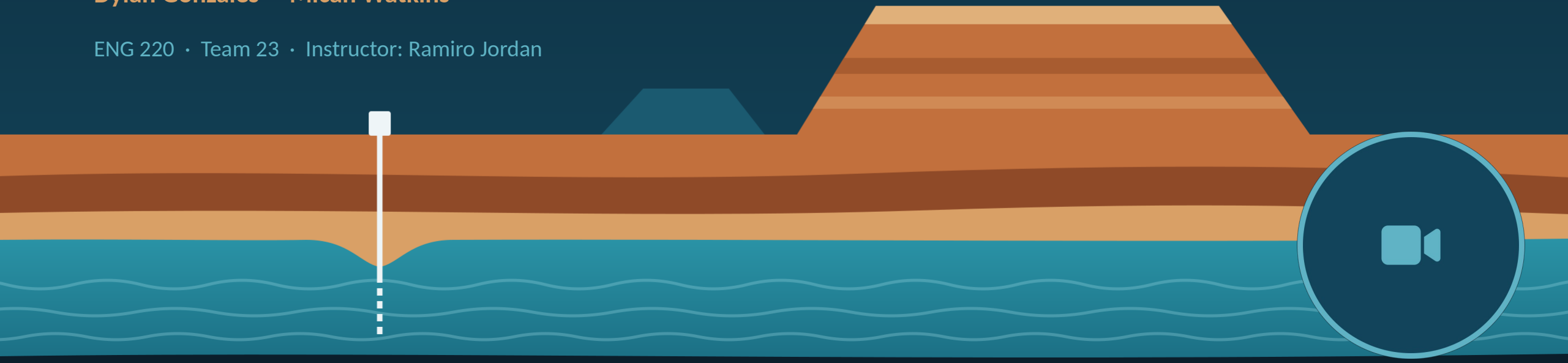
# Beneath the Mesa

Assessing Aquifer, Well, and Acequia Water Resources Across Sandoval County, New Mexico

**Petar Djordjevic (PM)** · **Meckrem Yousef** · **James Wauneka**
**Dylan Gonzales** · **Micah Watkins**

ENG 220 · Team 23 · Instructor: Ramiro Jordan



# The project at a glance

## What we're building

- A screening study of water resources across Sandoval County, with Cuba as the focus
- An interactive app, a written report, and an open GitHub repository

## What it does

- Brings together public well, aquifer, river, and acequia data
- Shows what's known about Cuba's groundwater, and where the data runs out

## Who benefits

- Cuba residents and village leaders
- Acequia communities and farmers
- Agencies deciding where to monitor next

## Peace engineering

Clear, shared information about a scarce resource helps communities make fair water decisions together.

# Background

- Where Sandoval County's water comes from: wells, rivers, and acequias
- Why the county spans several aquifer systems
- The concern: declining groundwater near Cuba
- What this project set out to check




*Figure caption and source*

# Research questions

MAIN QUESTION

**Write the project's main question in one sentence**

1. **Groundwater levels**
   Write the question in one sentence, using the wording from the proposal.

2. **Cuba's aquifers**
   Write the question in one sentence, using the wording from the proposal.

3. **Water quality and acequias**
   Write the question in one sentence, using the wording from the proposal.

# Deliverables

| Deliverable | What it is | Owner | Due |
|---|---|---|---|
| **Streamlit app** | Interactive maps and charts of the data | Meckrem | Nov 14 |
| **Final report** | Methods, findings, and figures in writing | Petar | Nov 14 |
| **Presentation slides** | This deck, presented in class | Petar | Nov 14 |
| **Individual videos** | Each member presents their own part | Everyone | Nov 15 |
| **Documentation site** | How each stage was done, with links to everything | Petar | Nov 16 |
| **GitHub repository** | Raw and cleaned data, scripts, and app code | Everyone | Nov 16 |

FINAL DUE

## Mon, Nov 16

**Nov 14**
App, report, and slides

**Nov 15**
Individual videos



# Data

## Groundwater levels

Source, years covered, and number of sites

## Aquifer tests

Source, years covered, and number of sites

## Water quality

Source, years covered, and number of sites

## Acequias

Source, years covered, and number of sites

*Full source links are on the last slide and in the project documentation.*



# Methods

## Collect

**Dylan Gonzales**

One or two lines on what this step did

## Clean

**James Wauneka**

One or two lines on what this step did

## Analyze

**Micah Watkins**

One or two lines on what this step did

## Present

**Meckrem Yousef**

One or two lines on what this step did

## Report

**Petar Djordjevic**

One or two lines on what this step did

**Tools** Python · pandas · Plotly · Streamlit · GitHub

# Data collection

## Dylan Gonzales

**Data acquisition and dataset justification**

**What I did**
The part of the project you were responsible for

**How I did it**
Tools, sources, and main steps

**What I learned**
A challenge you hit and how you handled it

**Screenshot of your work**

A data source page or the raw dataset

*Figure caption and source*

# Data cleaning

## James Wauneka

**Data cleaning and preprocessing**

**What I did**
The part of the project you were responsible for

**How I did it**
Tools, sources, and main steps

**What I learned**
A challenge you hit and how you handled it

**Screenshot of your work**

The cleaning script, or the data before and after

*Figure caption and source*

# Analysis and visualizations

## Micah Watkins

**Analysis code and visualizations**

**What I did**
The part of the project you were responsible for

**How I did it**
Tools, sources, and main steps

**What I learned**
A challenge you hit and how you handled it

**Screenshot of your work**

A chart from the analysis

*Figure caption and source*

# Streamlit app

## Meckrem Yousef

**GitHub workflow and the Streamlit app**

**What I did**
The part of the project you were responsible for

**How I did it**
Tools, sources, and main steps

**What I learned**
A challenge you hit and how you handled it

**Screenshot of your work**

A page of the live app

*Figure caption and source*

# Report and documentation

## Petar Djordjevic

**Project manager: report, slides, and video**

### What I did
The part of the project you were responsible for

### How I did it
Tools, sources, and main steps

### What I learned
A challenge you hit and how you handled it

**Screenshot of your work**

The documentation site or a page of the report

*Figure caption and source*

# Results

**Chart, map, or app screenshot**

Copy this slide for each result

*Figure caption and source*

## — ft

Key number and what it measures

### What this shows

Two or three sentences on the main pattern in the chart.

# Findings

### What the data show

- Measured results, with numbers

### What they suggest

- Likely explanations, stated as possibilities

### Too limited to tell

- Questions the data can't answer yet

**Bottom line**
One sentence that answers the main question



# Limitations

- Areas with few or no monitoring wells
- Short or irregular water-level records
- Data the project couldn't get
- What a stronger study would need

**Data coverage map**

Where monitoring is dense, sparse, or absent

*Figure caption and source*

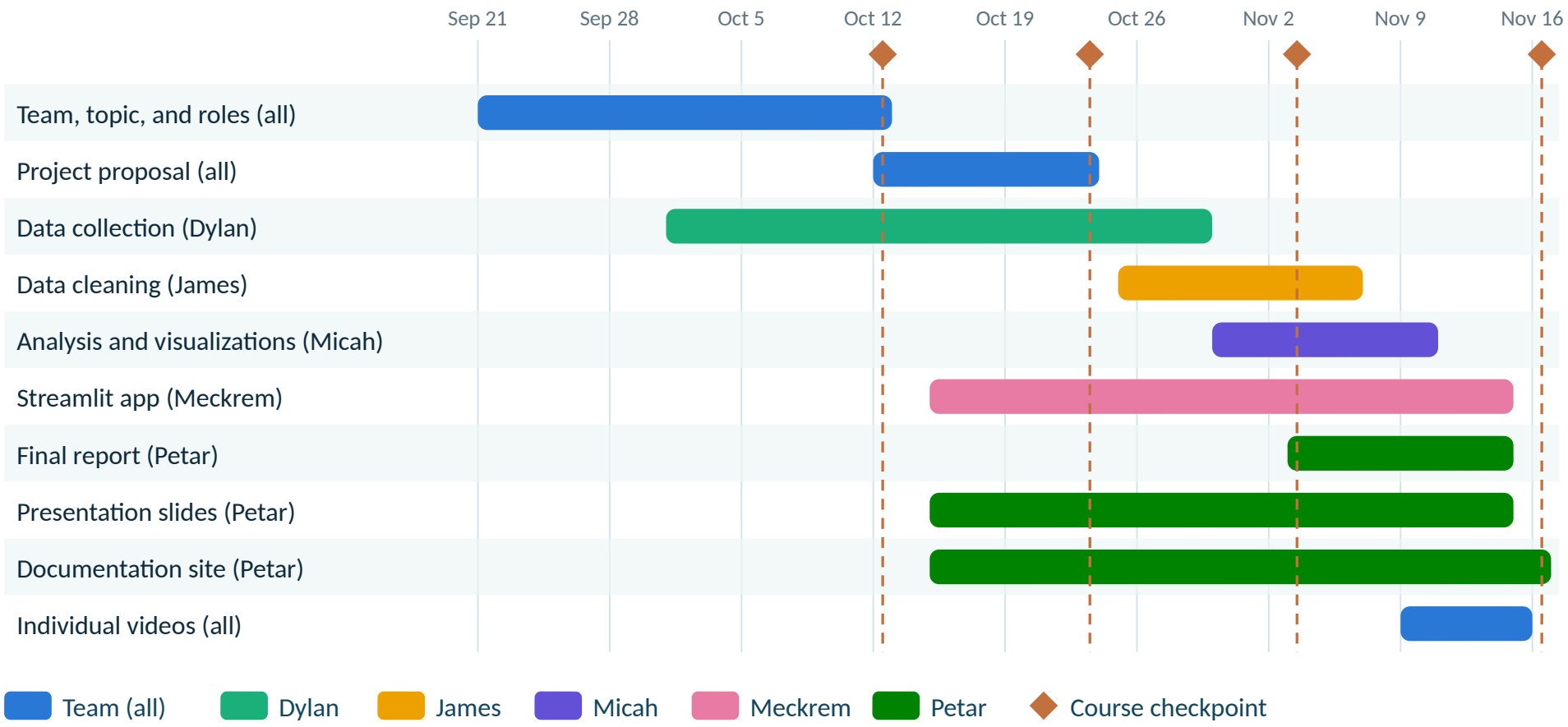
# Risks and how we reduce them

| Risk | How we reduce it |
| --- | --- |
| **No records from Cuba's own wells** | Ask the Village of Cuba and NM OSE for records; label every result as Cuba, county, or regional |
| **Sparse, uneven monitoring** | Map where coverage is thin, trend only wells with long records, and state the gaps |
| **Errors in the raw data** | Shared cleaning rules, a rerunnable script, and checks against the raw files |
| **Overstating conclusions** | Keep what the data show, suggest, and can't tell separate in every finding |
| **Tight final weeks** | Write the report and slides alongside the app, and track every deadline on the timeline |

## Guardrails from our proposal

- Treat the concern about Cuba's aquifer as a question, not a conclusion
- Keep Cuba as the main case study to stay on schedule
- Use public data anyone can recheck

# Timeline and milestones




## Milestones

- **Mon, Oct 12**
  First quarter · done
- **Fri, Oct 23**
  Midpoint · done
- **Tue, Nov 3**
  Three-quarter
- **Mon, Nov 16**
  Final due

# Explore the data, code, and app

**GitHub repository**
github.com/pdjordj11/eng_220_aquifer_project

**Streamlit app**
eng220aquiferproject.streamlit.app

**Documentation**
pdjordj11.github.io/eng_220_aquifer_project

**Datasets**
github.com/pdjordj11/eng_220_aquifer_project/tree/main/data

